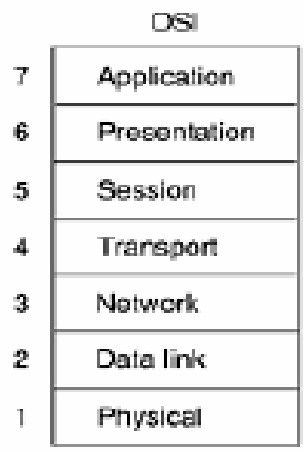
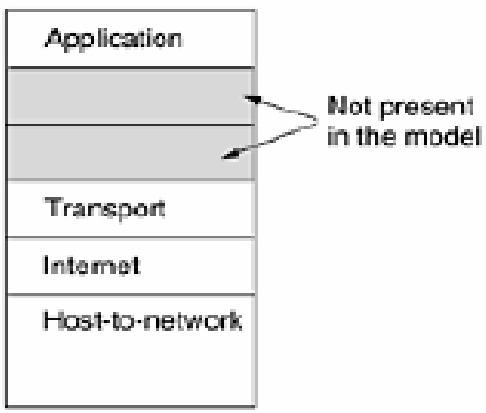
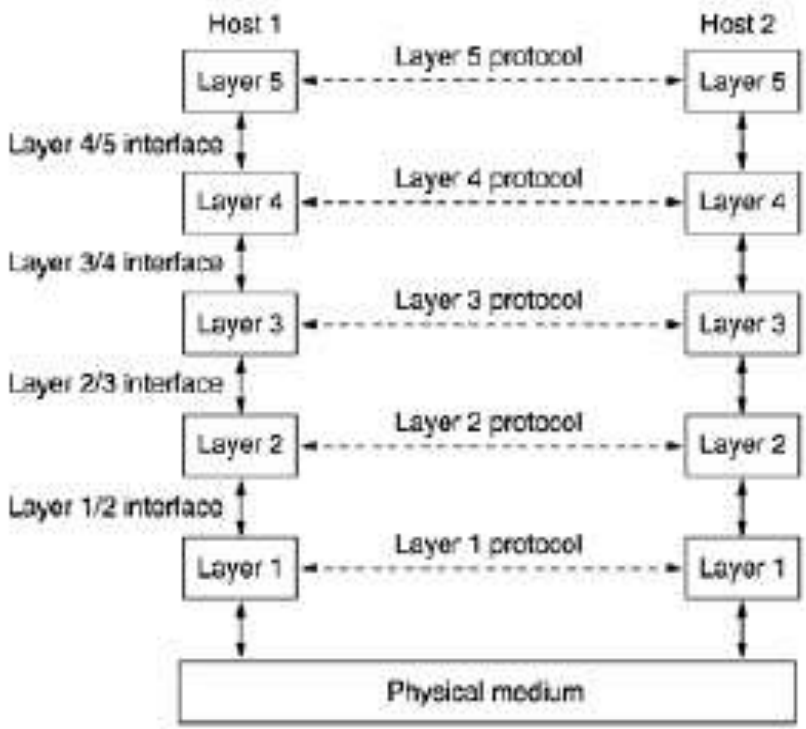
Differences between HDLC and PPP are(1) HDLC is cisco proprietary while PPP is an open standard.
(2) HDLC does not provide authentication while PPP provides authentication like PAP and CHAP.
(3) HDLC does not support compression while PPP support compression.
(4) HDLC supports synchronous networks only while PPP supports both synchronous and asynchronous networks.

***Contrast between OSI and TCP/IP reference models.

OSI Reference Model	TCP/IP Reference Model
OSI reference model is well defined reference model.	TCP/IP reference model is not well defined reference model.
OSI clearly distinguishes among service, interface and protocols.	TCP/IP does not clearly distinguish among service, interface and protocols.
Protocols in the OSI model are better hidden.	Protocols in the TCP/IP model are worse hidden as compared to OSI.
OSI model was devised before the protocols were invented.	TCP/IP model was described after the protocols come.
OSI has seven layers.	TCP/IP has four layers.
OSI supports both connection-less and connection-oriented communication in the network layer.	TCP/IP supports only connection-less communication in the network layer.
Supports only connection oriented communication in transport layer.	Supports both modes of communication in transport layer.
OSI is better for theoretical discussion and analysis.	TCP/IP is better for practical implementation.
	

Protocols Hierarchies:

To reduce their design complexity, most networks are organized as a stack of layers or levels, each one built upon the one below it. The number of layers, the name of each layer and the contents of each layer differ from network to network. In a sense, each layer is a kind of virtual machine, offering certain services to the layer above it. The following figure shows a five layer network.



***What are the Design issues for network layers?**

Some of the key design issues that occur in computer networks are present in several layers.

(i) Addressing:

Every layer needs a mechanism for identifying senders and receivers. Therefore, some form of addressing is needed in order to specify a specific location.

(ii) Error Control:

It is an important issue because physical communication circuits are not perfect. Many error detecting and error-correcting codes are known, but both ends of the connection must agree on which one is being used. In addition, the receiver must have some way of telling the sender which messages have been correctly received and which are not.

(iii) Flow Control:

See the question- Contrast between *Congestion Control* and *Flow Control*.

(iv) Congestion Control:

See the question- Contrast between *Congestion Control* and *Flow Control*.

(v) Fragmentation:

To solve congestion problem, large data are fragmented on the gateways. This process is known as fragmentation in design issues.

(vi) Routing:

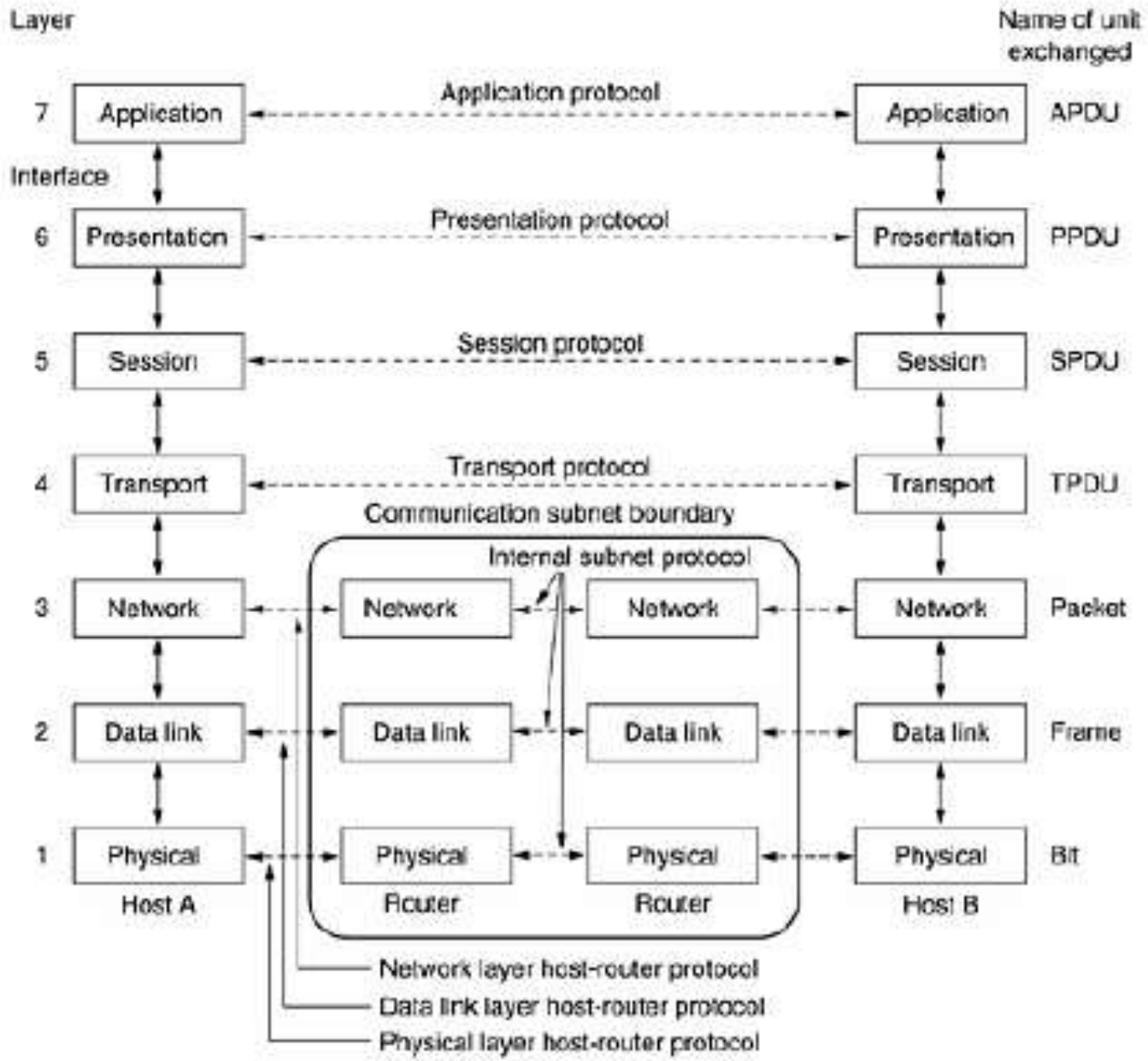
When there are multiple paths between source and destination, an appropriate route must be chosen. This decision making is called routing.

*Why layered approach is used for networking?

The layered approach is used in networking so that each layer can individually and efficiently performs their task in order to transmit a message.

*Describe briefly the OSI reference model.

The OSI Reference Model is so called because it deals with connecting open systems- that is, systems that are open for communication with other systems. The OSI has seven layers as shown in the following figure.


Layer
Name of unit exchanged
7
Application
Application protocol
Application
APDU
Interface
6
Presentation
Presentation protocol
Presentation
PPDU
5
Session
Session protocol
Session
SPDU
4
Transport
Transport protocol
Transport
TPDU
Communication subnet boundary
Internal subnet protocol
3
Network
Network
Network
Network
Packet
2
Data link
Data link
Data link
Data link
Frame
1
Physical
Physical
Physical
Physical
Bit
Host A
Router
Router
Host B
Network layer host-router protocol
Data link layer host-router protocol
Physical layer host-router protocol

The **physical layer** is concerned with transmitting raw bits over a communication channel.

The main task of **data link layer** is to control errors and make frames from.

The **network layer** is concerned with controlling the operation of the subnet.

The basic function of **transport layer** is to accept data from the session layer, split it up into smaller units if need be, pass these to the network layer and ensures that the pieces all arrive correctly at the other end.

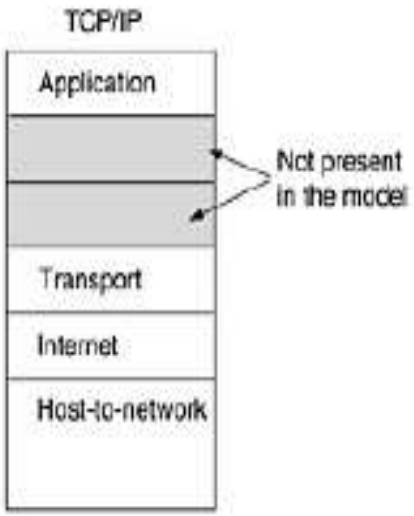
The **session layer** allows a user to establish session with another user on different machines.

The **presentation layer** is concerned with the syntax and services of the information transmitted.

The **application layer** allows a user to use virtual terminal software and efficient file transfer facilities.

***Describe the TCP/IP reference Model.**

The TCP/IP reference model has four layers instead of seven as compared to OSI reference model. The basic idea of TCP/IP reference model is described bellow.



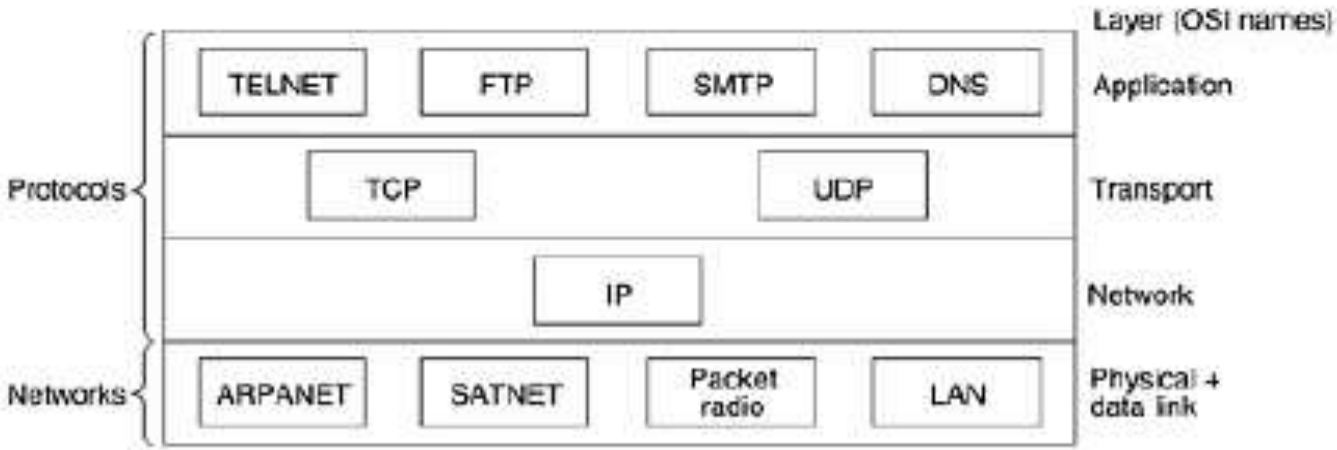
The **application layer** contains all the higher level protocols.

The **transport layer** allows peer entities on the source and destination hosts to carry on a conversation. There are two end-end protocols- UDP and TCP.

The **internet layer** defines IP and permits hosts to inject packets into any network and have them travel independently to the destination.

The **host-to-network layer** points out that the host has to connect to the network using some protocol.

***Draw the internal configuration of TCP/IP reference model.**



***What were the principles applied to arrive at layers?**

1. A layer should be created where a different level of abstraction is needed.
2. Each layer should perform a well defined function.
3. The function of each layer should be chosen with an eye toward defining internationally standardize protocols.
4. The layer boundaries should be chosen to minimize the information flow access the interfaces.
5. The number of layer should be large enough that distinct functions need not be thrown together in the same layer out of necessity, and small enough that the architecture does not become unwieldy.

What is the purpose of computer network?

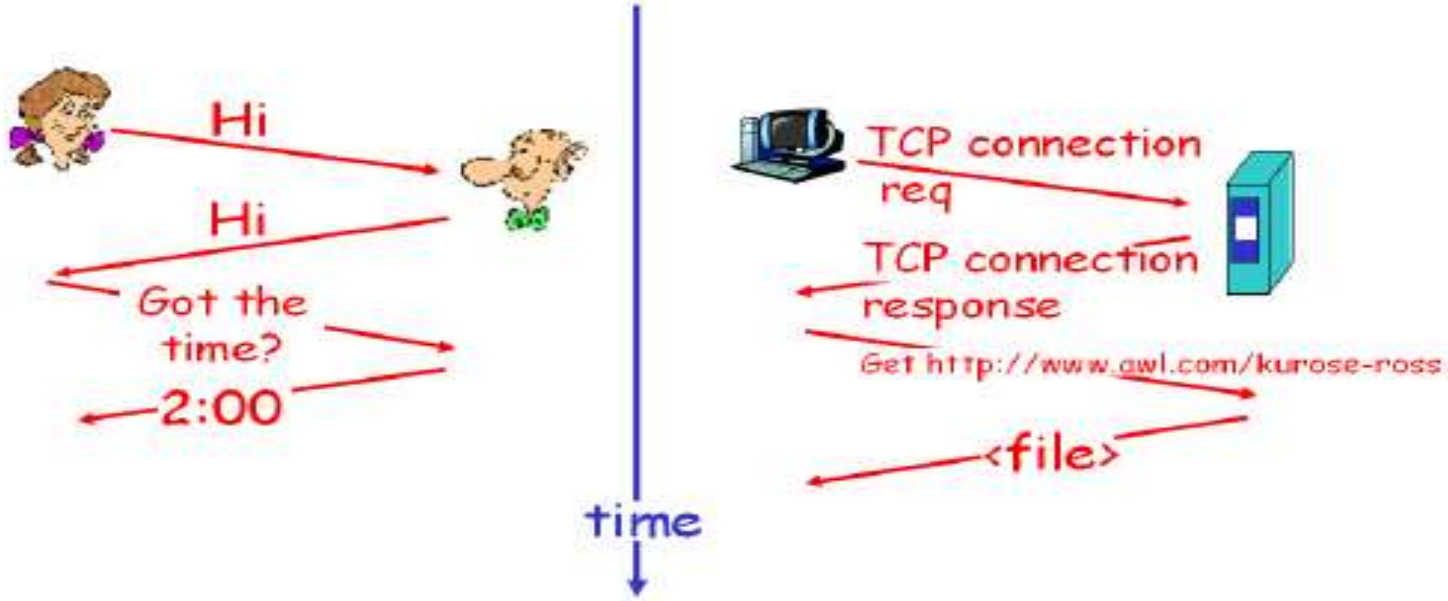
The purpose of computer network is to share information and resource like printers, databases etc. It also minimizes cost and increases productivity.

Why don't the intermediate routers defragment the packets in TCP/IP protocol?

The intermediate routers do not defragment the packets because of the fact the fragmented packets can go through paths that may need to further fragment the fragmented packets.

Give an analogy of human protocol and TCP/IP protocol.

The following figure shows an analogy between human protocol and TCP/IP protocol. While the human beings introduce themselves, a TCP/IP give connection requests and confirmation between two machines. Asking a question in human life is similar to asking for a certain service in TCP/IP. The following figure shows the idea in brief.



Why *session* and *presentation* layers are absent in TCP/IP?

TCP/IP was informally designed at first just to make up the necessity of communication. Further, modifying this, OSI model was developed.

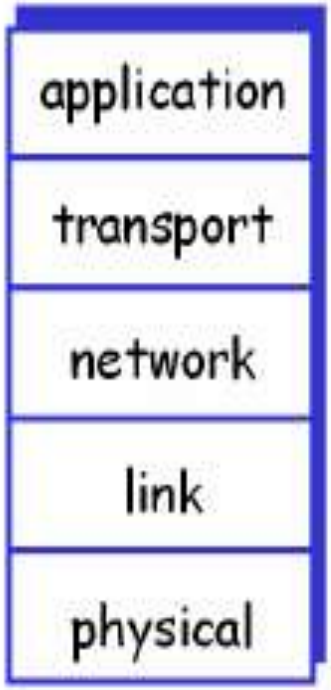
Why TCP is connection oriented?

Before establishing a connection, end-end receivers make handshaking. Also, receivers must send an acknowledgement that it received the respective packet.

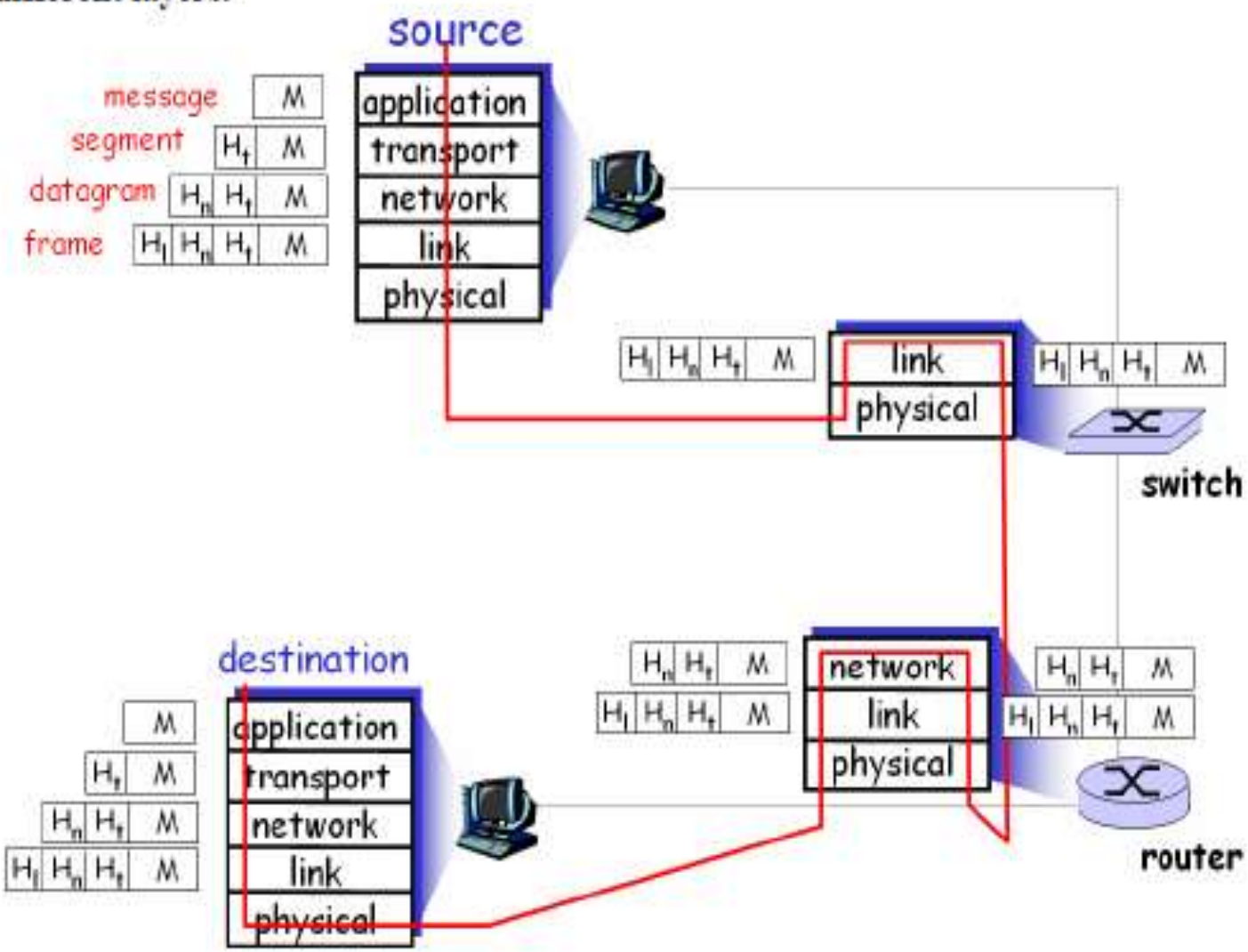
What is internet protocol stack? What are the tasks of each stack index?

Different layers designed for communication in internet is modeled as a stack which is called the internet protocol stack.

- application: supporting network applications
  - FTP, SMTP, STTP
- transport: host-host data transfer
  - TCP, UDP
- network: routing of datagrams from source to destination
  - IP, routing protocols
- link: data transfer between neighboring network elements
  - PPP, Ethernet
- physical: bits "on the wire"



Describe the process of encapsulation and decapsulation of a message through different layers.



Differentiate between packet switching and circuit switching.

In packet switching, there is no dedicated link between two hosts. This mode has the probability of occurrence of congestion. The example of packet switching is *internet*.

On the contrary, circuit switching mode requires a dedicated link between two terminals as done in telephone network. There cannot occur congestion.

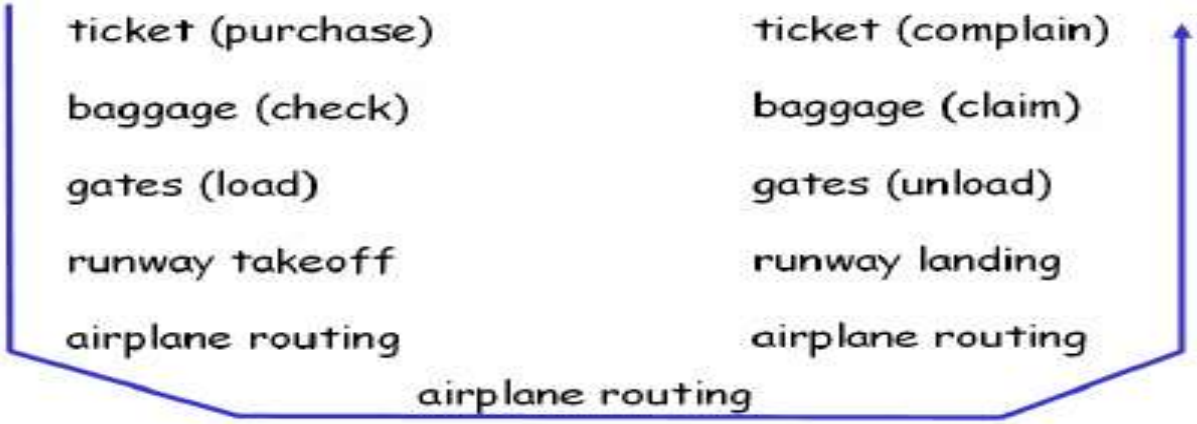
Why is layering required?

While dealing with the complex systems, layering gives us the following facilities.

-- explicit structure allows identification, relationship of complex system's pieces.
-- modularization eases maintenance, updating of system.
-- each layer implements a service.

Give an analogy of layers with airplane service.

In case of computer networks, we know when a segment or message needs to go from one machine to another, it goes through several layers like application, presentation, network, data-link, transport etc. It is analogous to a airplane service as shown in the following figure.



What are the limitations in case of data transfer in a computer network?

Though, in our simple concept, we know that the machine A just puts the bits on the wire, and machine B just takes them off. But there are several limitations in this case.

1. Communication circuits make errors occasionally.
2. They have only a finite data rate.
3. There is non-zero propagation delay.

What are the functions of data link layer?

The data link layer has a number of specific functions it can carry out. These functions include

1. Providing a well-defined service interface to the network layer.
2. Dealing with transmission errors.
3. Regulating the flow of data so that slow receivers are not swamped by fast senders.

What is the difference between Data Link Layer and Transport Layer?

Data Link Layer ensures whether the packet sent is received by the next nearest device while the *Transport Layer* ensures whether the data correctly reaches at the destination.

Discuss the relationship between packet and frames.

In order to accomplish the specified tasks for Data Link Layer, it gets packet from the network layer and encapsulates them into *frames* for transmission. Each frame contains a *frame header*, a *payload field* for holding the packet, and a *frame trailer*, as illustrated in the following figure.

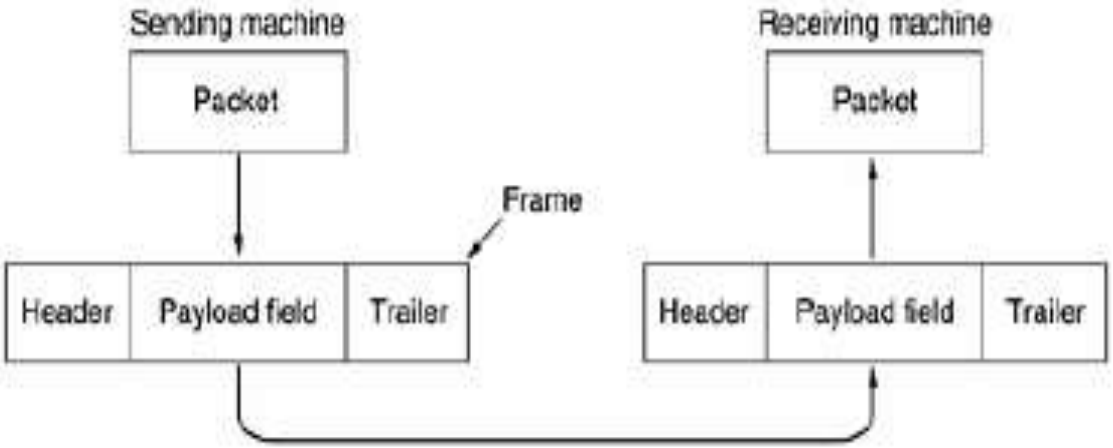


Fig.: Relationship between packets and frames.

What are the services of Data Link Layer that are provided to the Network Layer? Briefly Describe each of them.

The principal service is transferring data from the network layer on the source machine to the network layer on the destination machine. Other common services are-

1. Unacknowledged connectionless service:

Unacknowledged connectionless service consists of having the source machine send independent frames to the destination machine without having the destination machine acknowledge them. No logical connection is established beforehand or released afterward. If a frame is lost due to noise on the line, no attempt is made to detect the loss or recover from it in the data link layer. This class of service is appropriate when the error rate is very low so that recovery is left to higher layers. It is also appropriate for real-time traffic, such as voice, in which late data are worse than bad data. Most LANs use unacknowledged connectionless service in the data link layer.

2. Acknowledged connectionless service:

When this service is offered, there are still no logical connections used, but each frame sent is individually acknowledged. In this way, the sender knows whether a frame has arrived correctly. If it has not arrived within a specified time interval, it can be sent again. This service is useful over unreliable channels, such as wireless systems.

3. Acknowledged connection-oriented service:

With this service, the source and destination machines establish a connection before any data are transferred. Each frame sent over the connection is numbered, and the data link layer guarantees that each frame sent is indeed received. Furthermore, it guarantees that each frame is received exactly once and that all frames are received in the right order. With connectionless service, in contrast, it is conceivable that a lost acknowledgement causes a packet to be sent several times and thus received several times. Connection-oriented service, in contrast, provides the network layer processes with the equivalent of a reliable bit stream. However, it thus works in three phases- establish a connection, send some frames, releases the connection.

Illustrating the *Placement of Data Link Protocol*, discuss what does *Data Link Protocol* do?

Let us consider that a WAN subnet consisting of routers connected by point-to-point leased telephone lines. When a frame arrives at a router, the hardware checks it for errors, then passes the frame to the data link layer software. The data link layer software checks to see if this is the frame expected, and if so, gives the packet contained in the payload field to the routing software. The routing software then chooses the appropriate outgoing line and passes the packet back down to the data link layer software, which then transmits it. The flow over two routers is shown in the following figure.

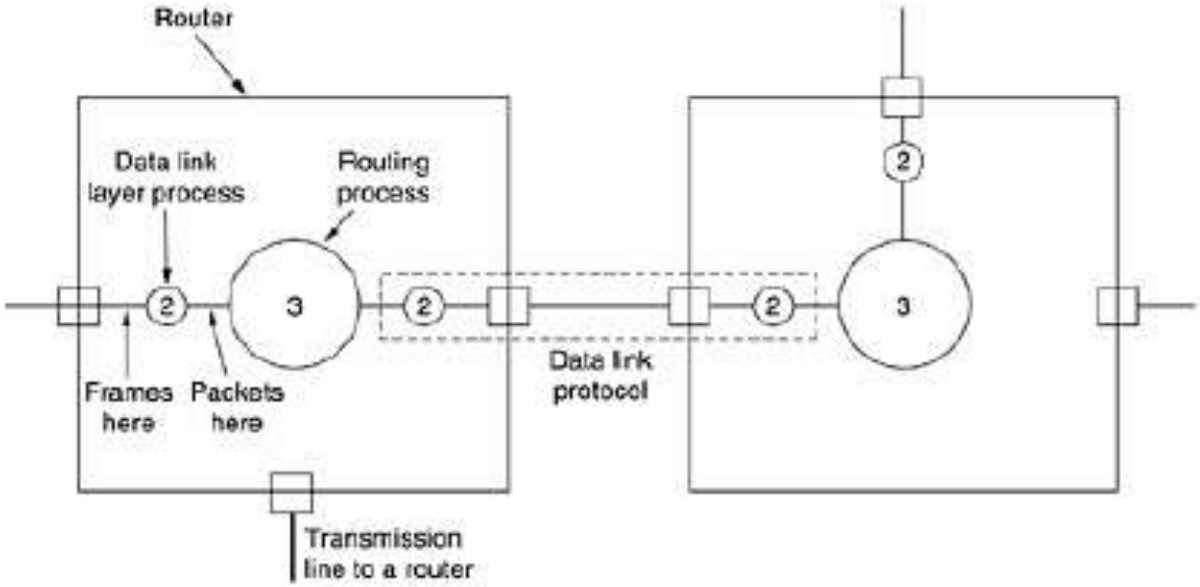


Fig.: Placement of *Data Link Protocol*

The routing code frequently wants the job done right. It does not want to be bothered with packets that got lost on the way. It is up to the data link protocol.

Differentiate between virtual communication and actual communication.

When a stream of bits is sent from one host to another, a communication path is required. In case of computer networks, it may be either virtual or actual. Physically, the path is actual and goes through the physical layer. But in case of theoretical consideration, it is to think a virtual path between two hosts from one network layer to another. The graphical depiction of virtual and actual communication is shown in the following figure.

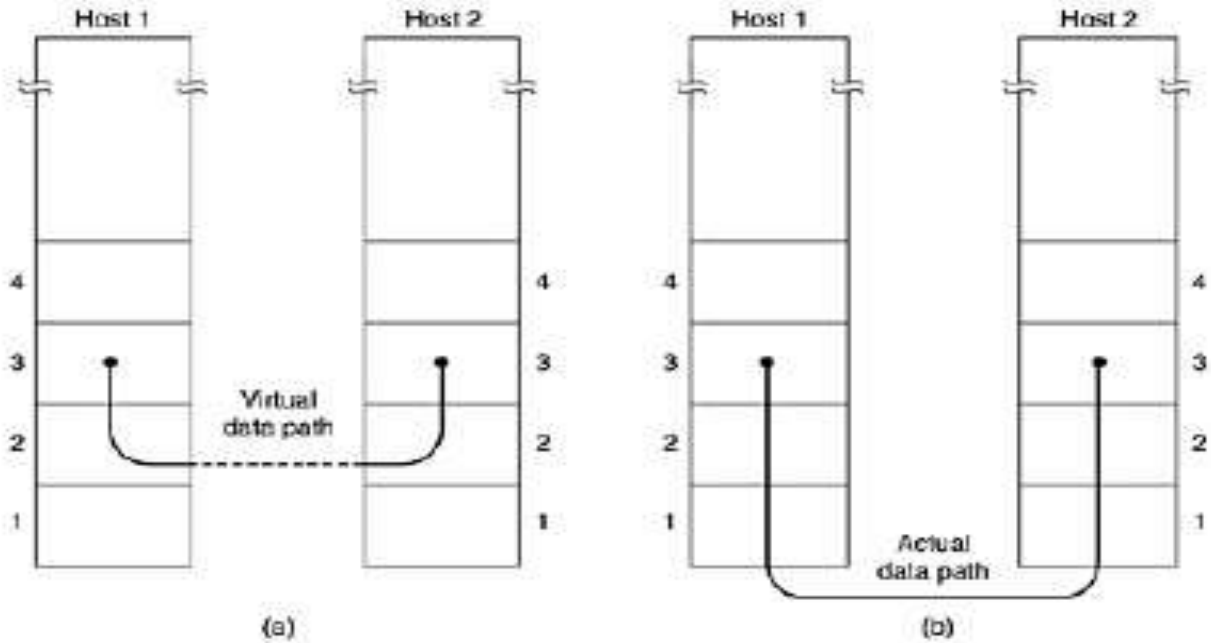


Fig.: (a) virtual communication and (b) actual communication

What do you mean by framing?

Data Link Layer receives packets from Network Layer, and delivers them to physical layer for transmission. But when these packets are received at the destination end, it is not guaranteed to be error free. To achieve considerable error control and detection mechanism, it is up to data link layer to break the packets and form frames that are better for handling errors. The process of making frames from packets is known as *framing*.

Discuss the *Receiving Techniques* of frames.

It is more difficult to receive frames in the form they were delivered due to different types of errors in the physical path. Thus, various receiving techniques are devised as categorized below.

1. Character Count
2. Flag Bytes with Byte Stuffing
3. Starting and Ending flags, with Bit Stuffing
4. Physical Layer coding violations

Character Count Technique:

This method uses a field in the header to specify the number of characters in the frame. When the data link layer at the destination sees the character count, it knows how many characters follow and hence where the end of the frame is. This technique is shown in the following figure for four frames of sizes 5, 5, 8, and 8 characters, respectively.

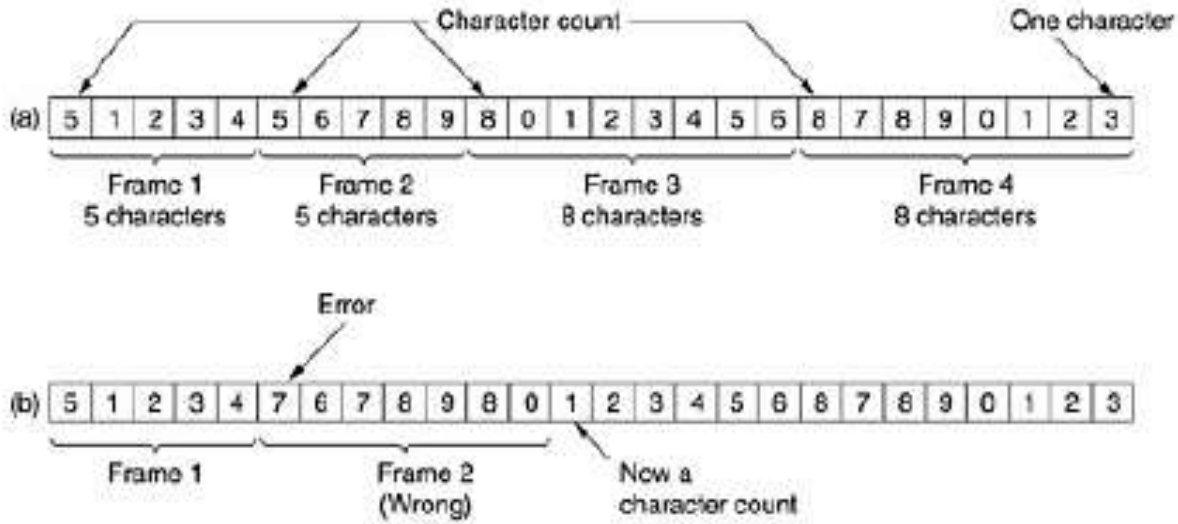


Fig.: Character Count Technique (a) without error (b) with error

The problem with this algorithm arises when in the transmission path, any character count is changed. For example, in the above figure, if the character count (5) of second frame becomes 7, the destination data link layer considers the next 7 bits as a frame and thus 1 becomes the next character count instead of 8.

Flag Bytes with Byte Stuffing:

In this method, no character count is defined at the beginning of the frame. Rather, an extra byte, called *flag byte*, is used at the beginning and at the ending of a frame so that the destination data link layer can understand the length of the frame. The technique is shown in the following figure (a).

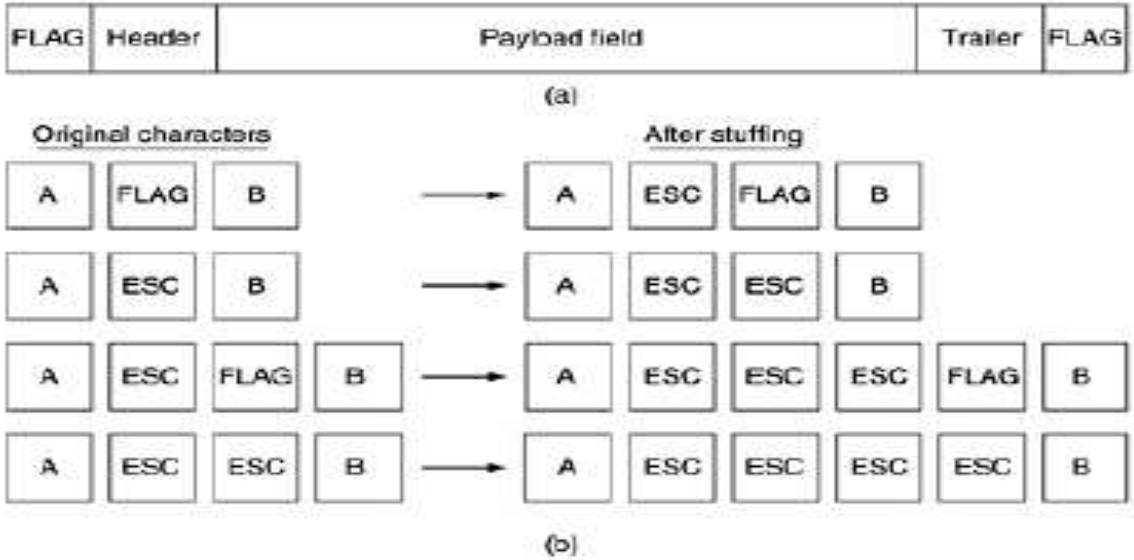


Fig.: (a) byte stuffing (b) escape byte

This algorithm may also present some error. Let us consider a situation when the bit pattern in the flag byte appears in the data filed. In this case, the destination data link layer will consider the bit pattern as ending flag and makes the end of the frame. To avoid such problem, a special byte, called ESC byte, is inserted before transmission as shown in figure (b).

A major disadvantage of this scheme is that, it strictly uses 8-bit bytes. But, not all the codes use 8-bit data. E.g. ASCII characters are 7 bits long.

Starting and Ending flags, with Bit Stuffing:

This techniques work like this. Each frame begins and ends with a special bit pattern, 01111110. Whenever the sender's data link layer encounters five consecutive 1s in the data, it automatically stuffs a 0 bit into the outgoing bit stream. When the receiver sees five consecutive incoming 1 bits, followed by a 0 bit, it automatically destuffs the 0 bit. If the user data contain the flag pattern, 01111110, this flag is transmitted as 011111010 but stored in the receiver's memory as 01111110.

(a) 0 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 0

(b) 0 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0 1 0 0 1 0

Stuffed bits

(c) 0 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 0

Fig.: (a) original data (b) data appearing on transmission line (c) received data

 # Physical Layer coding violations:

This method of framing is only applicable to networks in which the encoding on the physical medium contains some redundancy. For example, some LANs encode 1 bit of data by using 2 physical bits. Normally, a 1 bit is a high-low pair and a 0 bit is a low-high pair. The scheme means that every data bit has a transition in the middle, making it easy for the receiver to locate the bit boundaries. The combinations high-high and low-low are not used for data but are used for delimiting frames in some protocols.

****NOTE:: There are more two functions/design issues of Data Link Layer. One is Error Control and the other is Flow Control. Flow controlling can be accomplished by either **feedback-based flow control or rate-based flow control.***

****NOTE:: There are more two functions/design issues of Data Link Layer. One is Error Control and the other is Flow Control. Flow controlling can be accomplished by either feedback-based flow control or rate-based flow control.*

Define Burst Error.

A burst error is one in which a block of bits is considered erroneous where the first bit and last bit must be wrong.

What are the advantages and disadvantages of burst error over isolated single-bit errors?

Advantage: Computer data are always sent in blocks of bits. Suppose that the block size is 1000 bits and the error rate is 0.001 per bit. If errors were independent, most blocks would contain an error. If the errors came in bursts of 100 however, only one or two blocks in 100 would be affected, on average.

Disadvantage: Burst errors are much harder to correct than are isolated errors.

What are the basic strategies of dealing with errors?

There are two basic strategies for dealing with errors. For both the types, redundancy means adding a few extra bits with the original data-block. Thus,

n bit codeword = **m** data bits + **r** redundant/check bits

1. Error Correction: This method tells to include enough redundant information along with each block of data sent, to enable the receiver to deduce what the transmitted data must have been. On channels that are highly reliable, such as fiber, it is cheaper to use an error detecting code and just retransmit the occasional block found to be faulty. Error Correction is alternatively known as Forward-Error-Correction. The example of an error-correcting code - consider a code with only four valid code words:

0000000000, 0000011111, 1111100000, and 1111111111

This code has a distance 5, which means that it can correct double errors.

2. Error Detection: This method tells to include only enough redundancy to allow the receiver to deduce that an error occurred, but not which error, and have it request a retransmission. On channels such as wireless links that make many errors, it is better to add enough redundancy to each block for the receiver to be able to figure out what the original block was, rather than relying on a retransmission, which itself may be in error. The example of an error-detecting code - a single parity bit is appended to the data.

- Data - 1011010
- Even parity – 10110100
- Odd parity – 10110101

[NOTE:: To **detect** d errors, we need a d+1 code. Similarly, to **correct** d errors, we need 2d+1 code.]

Uses of Hamming Code to correct single-bit error:

To generate Hamming code, at 2^x positions i.e. at 1,2,4,8,16….positions we have to place the check bits and rest of the positions are filled by **m** message bits. Let the message be 1001000. Then, filling m positions by them, we get,

		1		0	0	1		0	0	0

Later, after putting all m bits, we have to break the **m** positions by a factor of 2 as follows-

3→1,2 5→1,4 6→2,4 7→1,2,4 9→1,8 10→ 2,8

11→1,2,8

Now, let us use 2^x positions to calculate-

1→ 3,5,7,9,11 [since 1 is in 3,5,7,9 and 11]

2→ 3,6,7,11

4→ 5,6,7

8→ 9,10,11

Now, for 1, (3,5,7,9,11) bit combinations are 10100. Since it is even parity, in position 1, we should place a **0**. Similarly, for 2, (3,6,7,11) bit combinations are 1010. It is also even parity, so, in position 2, we should place a **0**. The following figure completes the procedure. For any odd parity, place a **1**.

0	0	1	1	0	0	1	0	0	0	0

Uses of Hamming Code to correct burst error:

Hamming codes can only correct single errors. However, there is a trick that can be used to permit Hamming codes to correct burst errors. A sequence of k consecutive codewords is arranged as a matrix, one codeword per row. Normally, the data would be transmitted one codeword at a time, from left to right. To correct burst errors, the data should be transmitted one column at a time, starting with the leftmost column. When all k bits have been sent, the second column is sent, and so on, as indicated in the following figure. When the frame arrives at the receiver, the matrix is reconstructed, one column at a time. **If a burst error of length k occurs, at most 1 bit in each of the k codewords will have been affected**, but the Hamming code can correct one error per codeword, so the entire block can be restored.

Char.	ASCII	Check bits
H	1001000	00110010000
a	1100001	10111001001
m	1101101	11101010101
m	1101101	11101010101
i	1101001	01101011001
n	1101110	01101010110
g	1100111	01111001111
	0100000	10011000000
c	1100011	11111000011
o	1101111	10101011111
d	1100100	11111001100
e	1100101	00111000101

Order of bit transmission

Cyclic Redundancy Code (CRC) to detect errors:

CRC code is also known as polynomial code. In this case, the code is considered as a polynomial. For example, 110001 has 6 bits and thus represents a six-term polynomial with coefficients 1, 1, 0, 0, 0, and 1: x5 + x4 + x0. In this case, the sender and the receiver must agree upon a generator polynomial, G(x), in advance. Both the high- and low-order bits of the generator must be 1. The algorithm for computing checksum is given below.

1. Let r be the degree of G(x). Append r zero bits to the low-order end of the frame so it now contains m + r bits and corresponds to the polynomial x M r (x).

2. Divide the bit string corresponding to G(x) into the bit string corresponding to x M r (x), using modulo 2 division.

3. Subtract the remainder (which is always r or fewer bits) from the bit string corresponding to x M r (x) using modulo 2 subtraction. The result is the checksummed frame to be transmitted. Call it polynomial T(x).

What will be the transmitted frame if the given frame is 1101011011 using the generator G(x) = x4 + x + 1 when CRC code is used?

Frame : 1 1 0 1 0 1 1 0 1 1

Generator: 1 0 0 1 1

Message after 4 zero bits are appended: 1 1 0 1 0 1 1 0 1 1 0 0 0 0

```
                 1 1 0 0 0 0 1 0 1 0
10011 | 1 1 0 1 0 1 1 0 1 1 0 0 0 0
        1 0 0 1 1
          1 0 0 1 1
          1 0 0 1 1
            0 0 0 0 1
            0 0 0 0 0
              0 0 0 1 0
              0 0 0 0 0
                0 0 1 0 1
                0 0 0 0 0
                  0 1 0 1 1
                  0 0 0 0 0
                    1 0 1 1 0
                    1 0 0 1 1
                      0 1 0 1 0
                      0 0 0 0 0
                        1 0 1 0 0
                        1 0 0 1 1
                          0 1 1 1 0
                          0 0 0 0 0
                            1 1 1 0   <- Remainder
```

Briefly explain the HDLC and PPP protocol.

High Level Data Link Control (HDLC) protocol:

It is a bit oriented protocol, views the frame as a collection of bits. This protocol use bit stuffing for data transparency. It uses the following frame format. [all bit oriented protocols use this frame format]

Bits	8	8	8	≥0	16	8
	01111110	Address	Control	Data	Checksum	01111110

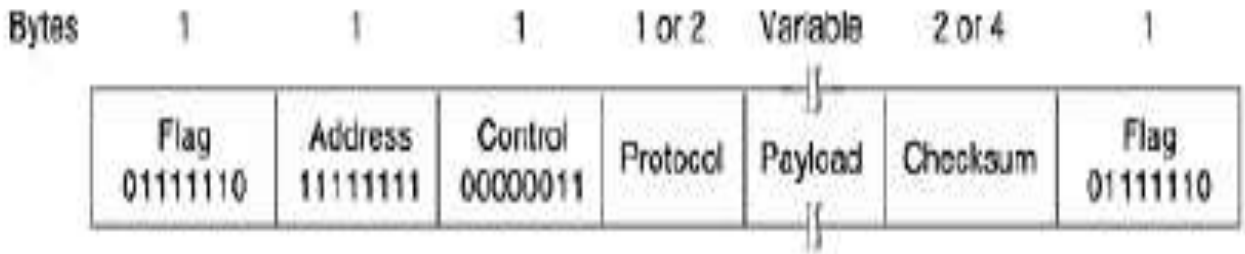
The Address field is primarily of importance on lines with multiple terminals, where it is used to identify one of the terminals. For point-to-point lines, it is sometimes used to distinguish commands from responses. The Control field is used for sequence numbers, acknowledgements, and other purposes. The Data field may contain any information. It may be arbitrarily long. The Checksum field is a cyclic redundancy code. [01111110 byte is a flag byte]

Advantages:

- HDLC provides both connection-oriented and connectionless service.
- HDLC can be used for point to multipoint connections.
- HDLC is the default encapsulation for serial interfaces on CISCO routers.
- HDLC frames can be transmitted over synchronous or asynchronous links.

Point-to-Point (PPP) protocol:

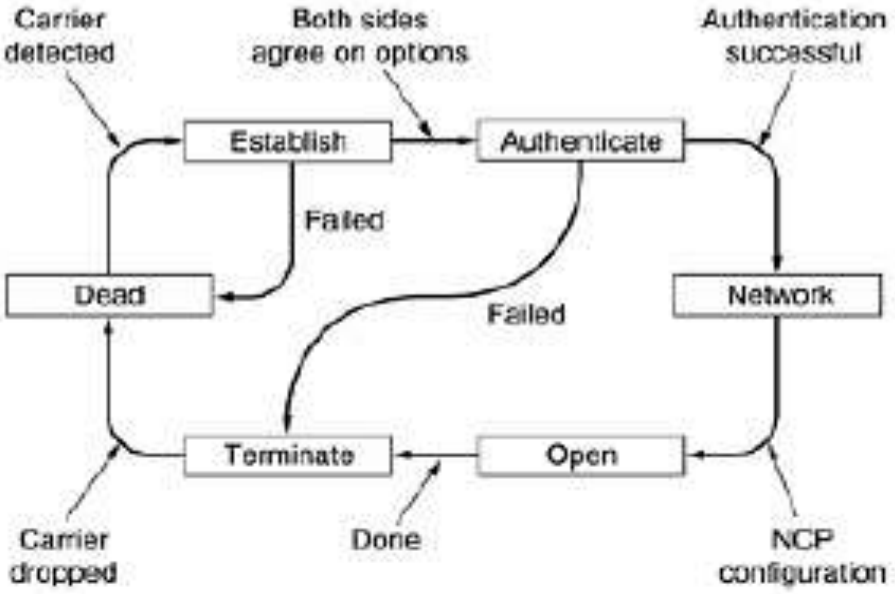
This protocol is character oriented rater than bit oriented as HDLC. It uses the following frame format.



Address field is always set to the binary value 11111111 to indicate that all stations are to accept the frame. Using this value avoids the issue of having to assign data link addresses. The default value of control field is 00000011. This value indicates an unnumbered frame. The job of payload field is to tell what kind of packet is in the Payload field. The Checksum field normally contains 2 bytes.

Simplified version of phase diagram for bringing a line up and down:

Such a diagram can describe a line is made up and down. The protocol starts with the line in the DEAD state, which means that no physical layer carrier is present and no physical layer connection exists. After physical connection is established, the line moves to ESTABLISH. At that point LCP option negotiation begins, which, if successful, leads to AUTHENTICATE. Now the two parties can check on each other's identities if desired. When the NETWORK phase is entered, the appropriate NCP protocol is invoked to configure the network layer. If the configuration is successful, OPEN is reached and data transport can take place. When data transport is finished, the line moves into the TERMINATE phase, and from there, back to DEAD when the carrier is dropped.



What do you mean by Collision Domain? How does a switch reduces the size of the collision domain while a hub increases it?

 # Collision Domain:

A collision domain is a physical network segment where data packets can "collide" with one another for being sent on a shared medium, in particular in the Ethernet networking protocol.

A Hub usually sends the packet to all the hosts connected to it when it receives a packet. The hosts that do not require that particular packet discard the packet while the hosts that need that packet accept it. Thus, connecting N hosts by a hub will make a collision domain of size N. In contrast, a switch can read the MAC address of a packet and can thus understand to which host a packet needs to be sent. The switch then sends the packet to that particular hosts and thus reduces the collision domain.

What is CSMA? How collision handled with the CSMA/CD?

 # CSMA:

CSMA stands for Carrier Sense Multiple Access. Protocols in which stations listen for a carrier and act accordingly are called carrier sense protocols. CSMA is of two types- *1-persistent* and *non-persistent*.

1-persistent CSMA:

When a station has data to send, it first listens to the channel to see if anyone else is transmitting at that moment. If the channel is busy, the station waits until it becomes idle. When the station detects an idle channel, it transmits a frame. If a collision occurs, the station waits a random amount of time and starts all over again. The protocol is called 1-persistent because the station transmits with a probability of 1 when it finds the channel idle.

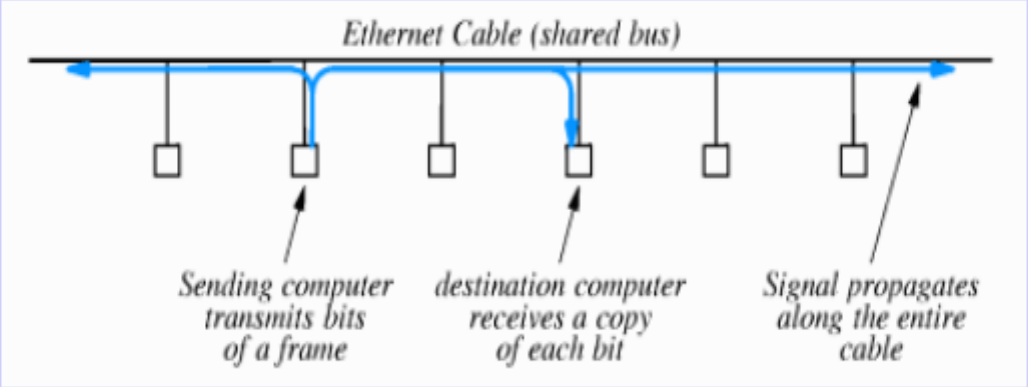
Non-persistent CSMA:

In this protocol, a conscious attempt is made to be less greedy than in the previous one. Before sending, a station senses the channel. If no one else is sending, the station begins doing so itself. However, if the channel is already in use, the station does not **continually** sense it for the purpose of seizing it immediately upon detecting the end of the previous transmission. Instead, it waits a random period of time and then repeats the algorithm. Consequently, this algorithm leads to better channel utilization but longer delays than 1-persistent CSMA.

Handling Collision with CSMA/CD:

CSMA/CD refers to CSMA with Collision Detection. In this approach, No station begins to transmit when it senses the channel busy. Stations abort their transmissions as soon as they detect a collision (i.e. abruptly). Quickly terminating damaged frames saves time and bandwidth.

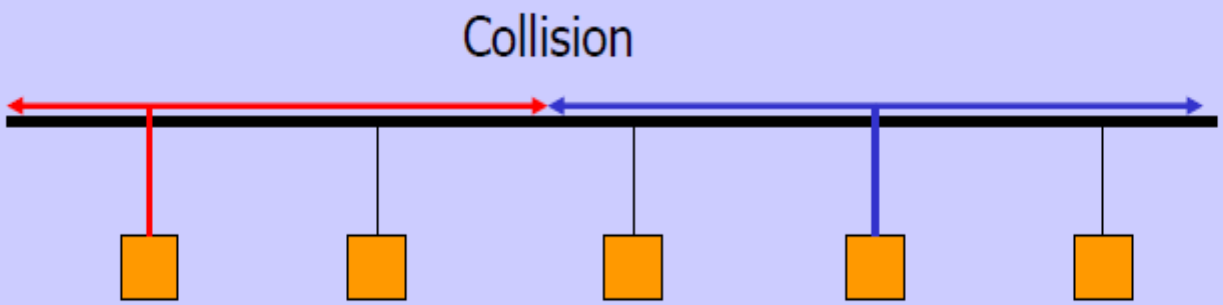
When to Start Transmit – CSMA



Animation

All computers connect to a bus but only one can transmit at a time.→Who can?
No central controller/manager when computers transmit on the bus!!
Ethernet employs CSMA to coordinate transmissions among multiple computers
CSMA (Carrier Sense with Multiple Access)
- **MA** (Multiple Access): multiple computers are attached, any can access bus
- **CS** (Carrier Sense): a computer listens to carrier before sending data
  → waits if medium/bus is busy (another computer is transmitting)
  → transmits if medium/bus is idle (no one transmits)

When to Start Transmit – CSMA/CD



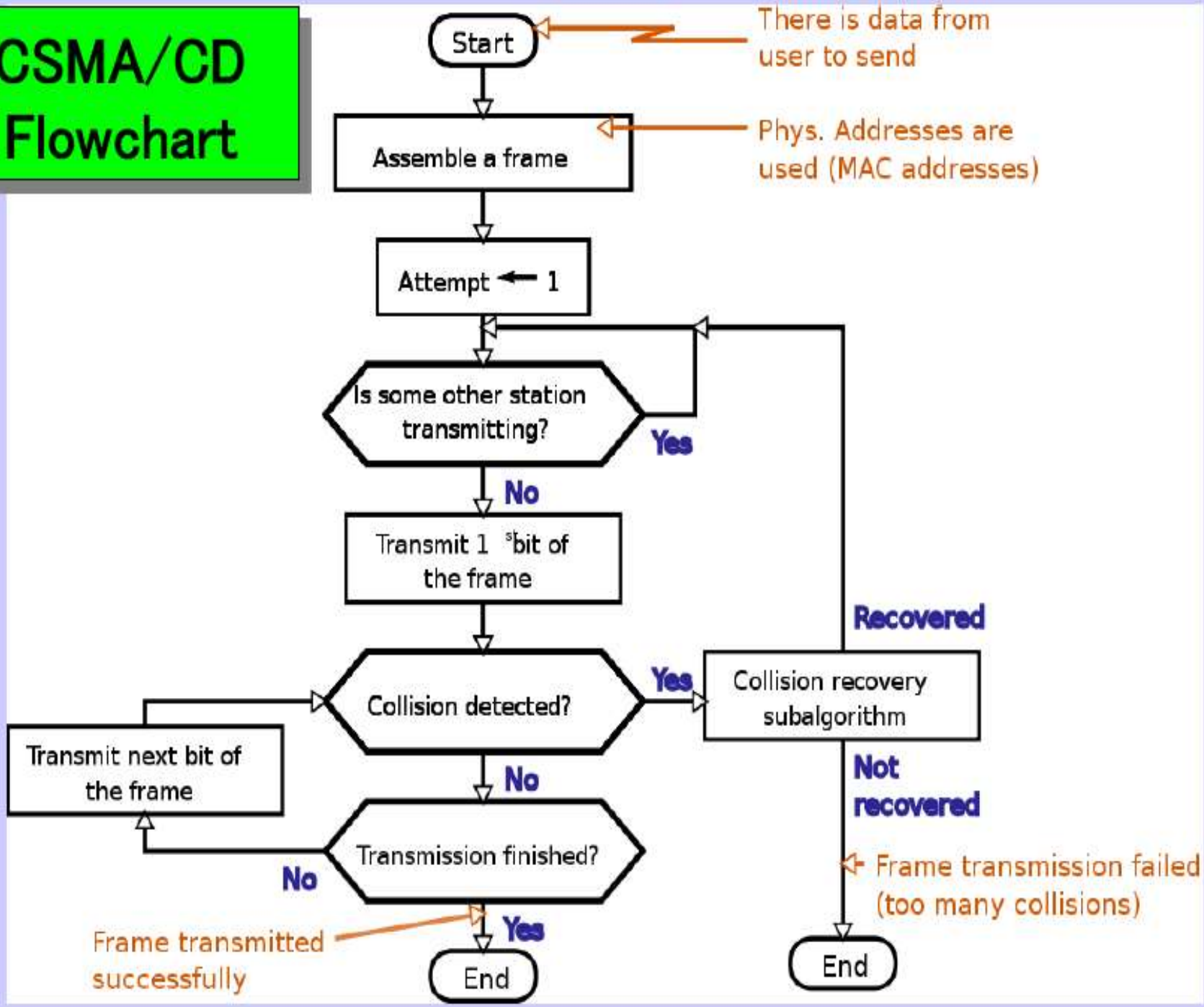
Animation

Animation

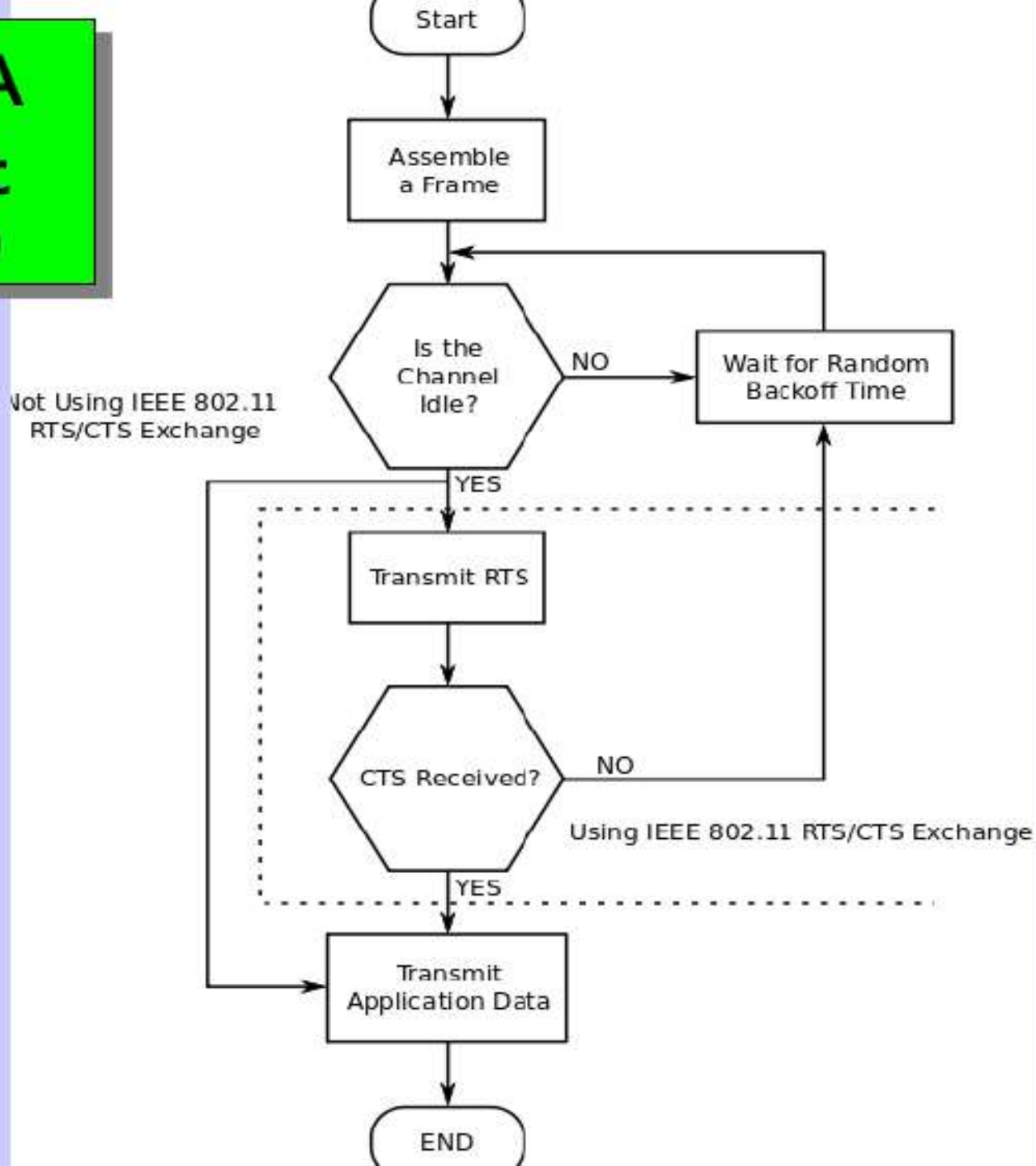
Even with CSMA, two computers may transmit simultaneously
- Both find it idle and begin transmitting in almost a same time
- Two signals exist in medium, interfere each other, call collision

CD (Carrier Detection)
- Monitor outgoing signal
- Stop transmissions after collision is detected
- Each waits random time in [0, S]
- Listen to the bus and transmit again if it is idle
- Wait random time in [0, 2S] if collisions is detected again
- Double waiting time range [0, 2^kS] when k+1 collisions detected
  → Called exponential backoff, very effective in practical

CSMA/CD Flowchart



CSMA/CA Flowchart (IEEE802.11)



Standard	Description
IEEE 802.11	The initial release of the standard capable of transmissions of 1 to 2 Mbps and operates in the 2.4 GHz band.
IEEE 802.11a	Capable of transmissions of up to 54 Mbps and operates in the 5 GHz band.
IEEE 802.11b	Introduced in 1999, 802.11b is capable of transmissions of up to 11 Mbps and operates in the 2.4 GHz band.
IEEE 802.11c	Defines wireless bridge operations
IEEE 802.11d	Defines standards for companies developing wireless products in different countries.
IEEE 802.11e	Defines enhancements to the 802.11 MAC for QoS.
IEEE 802.11f	Defines Inter Access Point Protocol (IAPP)
IEEE 802.11g	Capable of transmissions of up to 20 Mbps and operates in the 2.4, 3.6 and 5 GHz bands.
IEEE 802.11i	Improved encryption (WPA).
IEEE 802.11j	802.11 extension used in Japan.

The **Internet protocol suite** is the networking model and a set of communications protocols used for the Internet and similar networks. It is commonly known as **TCP/IP**, because its most important protocols, the Transmission Control Protocol (TCP) and the Internet Protocol (IP) were the first networking protocols defined in this standard. It is occasionally known as the **DoD model**, because the development of the networking model was funded by DARPA, an agency of the United States Department of Defense.

TCP/IP provides end-to-end connectivity specifying how data should be formatted, addressed, transmitted, routed and received at the destination. This functionality has been organized into four abstraction layers which are used to sort all related protocols according to the scope of networking involved.[1][2] From lowest to highest, the layers are the link layer, containing communication technologies for a single network segment (link), the internet layer, connecting independent networks, thus establishing internetworking, the transport layer handling host-to-host communication, and the application layer, which contains all protocols for specific data communications services on a process-to-process level.

The TCP/IP model and related protocols are maintained by the Internet Engineering Task Force (IETF).